



The population sizes of Mexican cities follow a power-law distribution

Héctor Saib Maravillo Gomez, Gilberto Calvillo Vives, and Erick Treviño Aguilar

Institute of Mathematics, National Autonomous University of Mexico, Cuernavaca, Morelos

ABSTRACT

Geography, for example because of the presence of rivers, ravines, or peaks, can subdivide a city. These subdivisions raise the question of identifying the area occupied by the city, of deciding whether or not they are aggregates of distinct built-up areas, and whether or not geographical separations are merely asperities in a certain continuum of built-up areas. The city as a union of administrative units allows for jurisdictional practices, but for public policy in health for example, identification by built-up areas is more operational. The study of urban populations thus requires that cities be circumscribed on objective criteria. Circumscribing a city requires knowledge of commuting flows, but in the absence of this piece of information, circumscribing it relies on the fact that it is made up of close built-up areas. This is reflected in the intersection of the convex envelopes of the spatial extent of these built-up areas. The algorithm treats coordinates of the vertices of polygons encompassing built-up areas provided by the Census Bureau for the United States or the National Institute of Statistics and Geography for Mexico. It allows for computing whether convex hulls of polygons intersect or not. If they do, then the built-up areas circumscribed by these polygons are part of the same city. The result is that cities now reflect the geographic extent of urban areas rather than their administrative areas. With this delineation method applied to Mexico's 2020 census data, the population sizes of urban areas with at least 2,126 inhabitants follow a power law, with exponent 0.954 (standard deviation = 0.016), whereas this is no longer the case when considering only the administrative extents of cities with more than 15,000 inhabitants.

KEYWORDS

City clustering algorithm; city system; distribution of population sizes of cities; intersection graph; power law; Zipf's law

JEL CLASSIFICATION

C21, R12, R14

1. Introduction

Contour lines of built-up areas are based on the spatial proximity and contiguity of buildings, blocks, and streets. Spatial databases (Matthews, 2020) have made it possible to delineate cities on the basis of built-up areas. The geography of places is not easily dealt with when circumscribing cities. For example, Harlem, East, and Hudson rivers, which run through New York City, impose their constraints on any grid used to map this city. The probability

distribution that characterizes population sizes of cities depends on the definition of the city used.

Circumscribing cities requires determining the minimum size at which the population is considered urban. This requires knowledge of commuting flows, which is seldom possible. Our method avoids specifying parameters or having to rely on administrative boundaries. We will show the robustness of our algorithm to geographical features and its capacity to cluster built-up areas based on the intersection of convex envelopes of built-up areas. The Mexican census of 2020 provides the population size by house and by block. This allows us to associate the population data as recorded in the census with each building and to calculate the size of the population of a city according to the boundaries assigned to it.

We present our algorithm for clustering cities by intersections of convex envelopes in [section 3.2](#). We shall test the algorithm with 2020 Mexican census data, which will allow us to validate that the population sizes of cities follow a power-law distribution after including the smallest recorded human settlements.

2. Background

2.1. Identifying a city

González-Val (2010) and Bee, Riccaboni, and Schiavo (2019) used the U.S. administrative units as given in the U.S. Census Bureau administrative base. However, neither the allocations of administrative units are the same across countries nor do the administrative boundaries correspond to the spatial distribution of the built environment. Both are the result of political history and ignore the economic relationships between places: a city may contain several municipalities or counties—for example, Mexico City’s built-up area includes 22 municipalities entirely and intersects with 31 others—conversely, several cities may be in the same county or municipality—for example, the municipality of Hermosillo, Sonora (15,720 km²), includes the cities of Hermosillo (86,000 inhabitants) and Miguel Aleman (39,000 inhabitants).

Garmestani, Allen, and Gallagher (2008) used “urbanized areas” defined by the U.S. Census Bureau as a collection of a central location with its urban fringe. Berry and Okulicz-Kozaryn (2012) used “economic areas,” which the Bureau of Economic Analysis of the United States Department of Commerce defined as one or more metropolises with the surrounding counties that are economically related by labor-force commuting.

Dijkstra and Poelman (2012) and Dijkstra, Poelman, and Veneri (2019) defined a “functional urban area” as a densely populated urban center with its surrounding area whose inhabitants participate in its labor market, thus

coming to work in the central location (these are not locations of firms connected by commercial links to firms located in the central location). To delimit these “functional urban areas,” Dijkstra, Poelman, and Veneri (2019) divided the study area into a square grid with cells of one kilometer of side. They knew the size of the population residing in each cell (an office area, for example, with no nightly inhabitants is ignored). They then identified an “urban center” consisting of contiguous cells each with at least 1,500 inhabitants per square kilometer and which together are populated by at least 50,000 people. They identified a city as one or more administrative local units in which at least half of the population resides in an urban center. The commuting area is a set of contiguous administrative local units with at least 15% of their employed residents working in the city. The functional urban area combines the city and its commuting zone.

Dijkstra, Poelman, and Veneri (2019) treated the case where an urban center spans two different cities. Schmidheiny and Suedekum (2015) used this method to deny that the population sizes of cities for the set of all European country members of the Organization for Economic Co-operation and Development follow a power-law distribution; Veneri (2016) showed that, in each of the countries of the Organization for Economic Co-operation and Development, the population sizes of functional urban areas follow a Zipf-law distribution (power law with exponent $\alpha = 1$). It is less clear for the population sizes of cities within their administrative boundaries.

To overcome the dependence on commuter flow data, Moreno-Monroy, Schiavina, and Veneri (2021) proposed to combine the previous functional definition with the distance of cells to the nearest urban center in order to estimate the probability that a cell outside an urban center belongs to a metropolis. Holmes and Lee (2010) objected that metropolises aggregate administrative units that include rural areas and therefore mix heterogeneous settlement entities. Because our method does not consider administrative units, it allows for consideration of small towns and human settlements.

Rozenfeld et al. (2008)’s algorithm ignores administrative subdivisions and center-periphery models. It is based on a partition of the territory into a fine square mesh where each cell is associated with the size of its resident population. Adjacent cells that are not empty of inhabitants are iteratively grouped together, so that each group thus formed defines a city. Rozenfeld et al. (2011) proposed a continuous-space version of this algorithm, based on the spatial coordinates of census data. The points are then grouped according to a minimum neighborhood distance rule. The threshold of at least one inhabitant becomes a density threshold. This algorithm has variants: Oliveira et al. (2018) added criteria that depend on the area measure of the clusters at each iteration; Cao et al. (2020) used a percolation criterion to set the minimum density threshold used in circumscribing urban areas; Xu et al. (2021) delineated “urban settlements” on a rank criterion from remotely sensed data of

built-up areas and water bodies; Chen et al. (2022) grouped urban areas if their distance was less than a predetermined threshold. They showed that these groups follow an exponential distribution within each of the 24 Chinese cities they studied.

Alternatively, Zhou et al. (2015), Bergs (2018), and Wu, Zhao, and Jiang (2018) used night-light satellite images from the defense meteorological satellite program to circumscribe cities. Yu et al. (2014) and Wang et al. (2022) clustered urban areas extracted from night-light satellite images using graph partitioning techniques to form urban clusters. Tannier et al. (2011) and Tannier and Thomas (2013), based on satellite images, deduced morphological boundaries of urban agglomerations in France and Belgium by first producing a vector image of the Minkowski dilation of each building appearing on the map. This consists of overlaying each frame image with a larger dilated polygon from the original image. The algorithm groups intersect dilated polygons. A log-log curve then relates the total number of clusters thus created to the width of the dilation, revealing an optimal dilation width. Yin et al. (2017) used a network of Twitter user interactions and the network clustering algorithm “infomap,” which allows for segregating the nodes of a graph to form well-connected subgraphs, to circumscribe cities in Great Britain. Ren, Jiang, and Seipel (2019) identified urban settlements in the United States by combining geospatial information from Twitter users and the mapping of buildings. In order to circumscribe cities that use data from street networks and the mapping of building, Montero, Tannier, and Thomas (2021) compared the method of identification of “natural cities” by clustering street nodes and building centroids (Jiang and Jia, 2011), the “MorphoLim” method based on fractals (Tannier and Thomas, 2013), and the hierarchical percolation (Masucci, 2015). The city boundaries provided by these methods, although based on the morphology of built-up areas, are still dependent on geographical features. The authors cited in this paragraph used the morphology of built-up areas without considering administrative or economic criteria.

2.2. The case of Mexican cities

2.2.1. Distribution of the population sizes of cities

Auerbach (1913) showed that, empirically, the products of population sizes of cities times the ranks of these population sizes approximated a country-dependent constant. He argued that this constant could be used as a measure of a country’s urban concentration instead of the population density. For example, although India has a high population density, the majority resides in the countryside or in small towns. India and Great Britain had comparable population densities, based on censuses conducted between the years 1909 and 1912, but India’s population was mostly rural, while Great Britain’s population was mostly urban. Auerbach’s measure of

concentration, based on the products of population sizes of cities times population ranks, was eight times higher in Great Britain than in India and, thus, reflects this difference. Zipf (1949) showed that both the frequency of words in languages and the population sizes of cities also follow a power-law distribution. Mandelbrot (1966) noticed that the power-law distribution of words in a text could reflect the evolutionary optimization of communication.

In the microeconomic model of Hsu (2012) based on the assumption that cities of different sizes have different economic functions, the power-law distribution of the population sizes of cities results from the “regularly varying” distribution of scale economies. A measurable positive function g is *regularly varying* (Hsu, 2012) with index $a \in \mathbb{R}$ if and only if, for any $\lambda > 0$,

$$\lim_{x \rightarrow \infty} \frac{g(\lambda x)}{g(x)} = \lambda^a, \quad (1)$$

and a *scale economy* is an economy where the cost of production decreases with the marginal increase in the quantity produced. Here, large cities would offer more commodities than small cities. For Verbavatz and Barthelemy (2020), city growth is dominated by large interurban migratory flows rather than by the average population growth rate. The population size of a city is a random variable X , which follows a power-law distribution if its probability density function p is such that

$$p(x) \propto x^{-(\alpha+1)} \quad (2)$$

for $\alpha > 0$ and where x is the population size of the city. The complementary cumulative distribution function of X is

$$P(x) := \Pr(X \geq x). \quad (3)$$

Newman (2005) and Clauset, Shalizi, and Newman (2009) have shown that the complementary cumulative distribution function $P(x)$ is proportional to $x^{-\alpha}$ if the random variable X follows a power law. If x tends to 0, then $p(x)$ converges to infinity. Therefore, the power law holds only above a cutoff point $x_{\min}$.

If the exponent α is large enough, the population size tends to be evenly distributed among cities, and if α tends to 0, then the population size distribution is inegalitarian, with a few megacities and many small cities (Verbavatz and Barthelemy, 2020). Zipf’s law is the power-law distribution with exponent $\alpha = 1$.

We follow Clauset, Shalizi, and Newman (2009) by estimating the value of the exponent α with the maximum likelihood and the cutoff threshold $x_{\min}$ with the Kolmogorov–Smirnov distance, assessing the goodness of fit to the empirical data, and then comparing this power-law fit to those obtained with the log-normal and the exponential distributions using the likelihood ratio

test. Alstott, Bullmore, and Plenz (2014) programmed these three steps in Python and Gillespie (2015) in R.

Rosen and Resnick (1980) rejected Zipf's law for 36 countries. For Mexico, their ordinary least squares regression yields an exponent $\hat{\alpha} = 1.153$ (standard deviation $SD = 0.020$). Soo (2005) also rejected Zipf's law for 53 of the 73 countries examined. For Mexico, they used a sample of 162 cities for the year 2000 and found $\hat{\alpha} = 0.9725$ (SD not given) by ordinary least squares. Veneri (2016), using data from "functional urban areas" in 29 OECD countries, rejected the power-law distribution for the population sizes of cities for only Mexico, Poland, and Spain. For Mexico, they rejected the power-law distribution for 77 "functional urban areas" but not for the 76 official cities counting more than 50,000 inhabitants as reported in the year 2001. Oliveira et al. (2018) successfully fitted Zipf's law for 145 countries comprising at least 10 cities, with mean $\hat{\alpha} = 0.98$ ($SD = 0.30$). They estimated the exponent $\hat{\alpha} = 0.726$ ($SD = 0.043$) for Mexico by maximum likelihood. Puente-Ajovín, Ramos, and Sanz-Gracia (2020), using data from Mexican municipalities in the year 2000, found that, according to the Bayesian information criterion, the bi-lognormal distribution provides a better fit to the 70-country distribution of population sizes of cities than the lognormal, log-logistic, double Pareto lognormal, bi-logistic, tri-lognormal, and tri-logistic distributions. Xu et al. (2021), based on the areas of urban settlements in 47 countries in the year 2018, rejected that the population sizes of these settlements follow a power law only for Belgium, Russia, Belarus, and Vietnam. For the case of Mexico, their data concern 171 urban settlements. They estimated $\hat{\alpha} = 1.76$ ($SD = 0.06$). Urzúa (2001), using data from the 1960 to the 1990 decennial censuses of Mexico's 81 official cities—only two cities had less than 10,000 inhabitants in 1970, and in 1990 all cities considered had more than 30,000 inhabitants—rejected Zipf's law for all years considered. Pérez-Campuzano, Guzmán-Vargas, and Angulo-Brown (2015), on 20th-century decennial census data from 33 cities in 1900 to 350 cities in 2000 with 15,000 or more inhabitants, estimated a steady decline from each decennial census to the next from 1900 to 2000. For example, they found $\hat{\alpha} = 1.329$ in 1990 ($SD = 0.231$) and 0.842 ($SD = 0.044$) in 2000 by maximum likelihood. These studies ignore urban settlements with populations of less than 15,000, a gap that we will fill.

2.3. Power law and coherence

Although the population sizes of cities for each European country follow a power-law distribution, this is no longer true for Europe as a whole (Cristelli, Batty, and Pietronero, 2012). Likewise, in the United States, the power-law distribution that describes the population sizes of U.S. cities for the country as a whole is no longer valid within each state (in contrast to European states) (Cristelli, Batty, and Pietronero, 2012). Cristelli, Batty, and

Pietronero (2012) hold economic and social integration as the reason for this difference, which they call “coherence with the emergence of a power law.” We expect that Mexico, because of its relative cultural and economic homogeneity, presents this property of coherence.

3. Method and data

3.1. Data of localities and metropolises in Mexico

The Mexican National Institute of Statistics and Geography (INEGI) is in charge of organizing the population and housing census every year multiple of 10. The basic geographic units for conducting the census are blocks and localities. A *block* is an area that covers a set of houses and buildings surrounded by streets. *Localities* are places “occupied with one or more buildings used as dwellings, which may or may not be inhabited [...], recognized by a name given by some legal provision or custom.”¹ These two units are defined by the coordinates and physical characteristics recorded by census agents and are periodically updated with satellite images. A locality appears on the map as a polygon that covers a set of adjacent blocks. For reasons of comparability with administrative units, the polygons of the localities are constrained by their administrative boundaries, which are often dependent on geographical or historical characteristics. Figure 1 shows an example. The official threshold between rural and urban localities is 2,500 inhabitants. The 2020 population census counted 126,014,024 inhabitants in Mexico, of which 96.59% resided in 4,911 urban localities and 45,037 rural localities. The rest of the population resided in scattered rural localities without any block. The National Institute of Statistics and Geography has published open data² on the boundaries and population sizes of urban localities—hence 2,500 inhabitants or more—since the 2000 census. In the 2020 census, it added the coordinates of polygons for rural areas, too. This piece of information allowed us to estimate the lower bound where the power law begins.

Since 2018, the Ministry of Agriculture, Land, and Urban Development (SEDATU in Spanish), in collaboration with the National Population Council (CONAPO in Spanish), has been developing the national urban system, with the objective of supporting public policies for urban planning. It includes 401 cities, of which 74 are metropolises. The latter are formed by grouping municipalities, including rural ones if any. The remaining official cities are defined solely from urban locality data. The delimitation of metropolises, according to planning programs and territorial ordering, follows the method of SEDATU-CONAPO-INEGI (2018b) consisting of first defining a set of central municipalities according to

¹<https://www.inegi.org.mx/app/glosario/default.html?p=localidades>.

²<https://www.inegi.org.mx/temas/mg/#Descargas>.

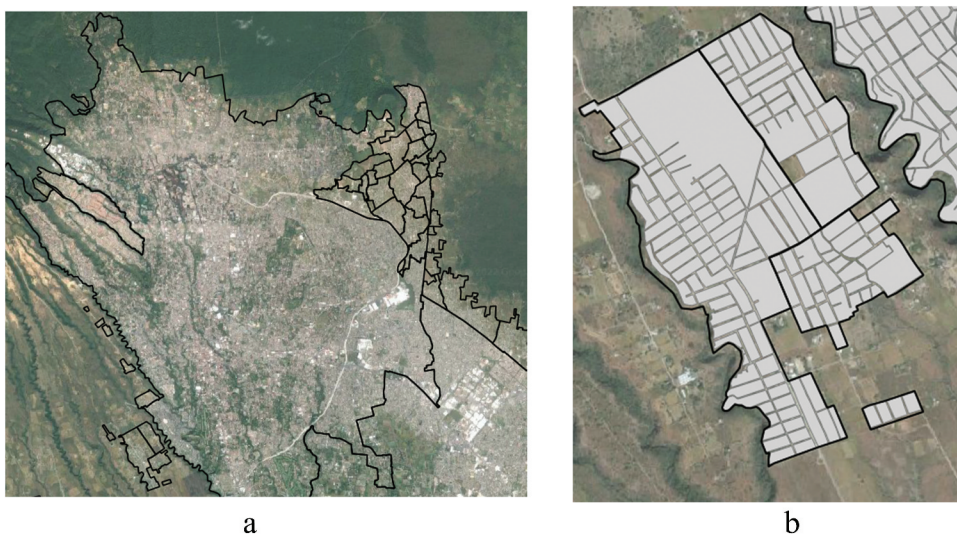


Figure 1. City of Cuernavaca, Morelos. Map (a) is a satellite image of the city of Cuernavaca and locality boundaries with black lines. Map (b) is a close-up of three localities on the outskirts of the city of Cuernavaca. The gray polygons represent the blocks. The three localities are not adjacent to the center of Cuernavaca because of the mountain slope.

demographic and geographic criteria and second defining a set of external municipalities in case of contiguity with the central municipalities and according to economic criteria (its most populated location within 15 km of the central city; at least 15% of its labor force is employed in the central city; less than 25% of its labor force is employed in the primary sector; and at least 20 residents per hectare) and urban policy criteria (municipalities outside a metropolis must be considered in the metropolis declaration, the metropolis management program, and the national urban development program). Finally, a metropolis is the union of its central municipalities with their external municipalities.

3.2. City clustering algorithm by intersecting convex envelopes

Our algorithm of “clustering cities by intersection of convex envelopes” is based on the morphology of the built-up environment. It reflects spatial continuity, restores the area of influence of cities, and groups built-up areas while avoiding the introduction of subjective parameters or the imposition of a center-periphery scheme.

Unlike the algorithm by Rozenfeld et al. (2008), which groups only cells of a square mesh, it joins adjacent building polygons. These polygons cover sets of adjacent blocks, so that, for example, two blocks along the same street are in the same polygon. To overcome the difficulty of geographical barriers, such as a river, a mountain slope, or unbuilt

areas, the algorithm groups polygons that are sufficiently close, understood as the fact that the convex envelopes of the polygons have a non-empty intersection. If the polygons correspond to the populated cells of a square mesh, then the algorithm is equivalent to the city clustering algorithm by Rozenfeld et al. (2008).

The 2020 Mexican population census provides the coordinates of the blocks and streets of each city. These data are not unique to Mexico: this is the case in America for the Brazilian Institute of Geography and Statistics, the National Statistical Office of Canada, the National Statistical Institute of Chile, the Census Bureau of the United States, and the National Statistical Institute of Uruguay. Figure 2 presents the algorithm for clustering cities by intersection of convex envelopes:

- Step 1: compute the convex envelope K_i on the map for the polygon i of each built-up area in the plane. The convex hull of a polygon X with vertices $x_1, x_2, \dots, x_k$ is defined as

$$\text{conv}(X) = \left\{ \sum_{i=1}^k \alpha_i x_i \mid \alpha_i \in [0, 1], \sum_{i=1}^k \alpha_i = 1 \right\}. \quad (4)$$

The set of convex envelopes is $K = \{K_1, \dots, K_n\}$.

- Step 2: the set K is also the set of nodes of the intersection graph $G = (K, E)$, where E is the set of edges. An edge joins two vertices K_i and K_j if and only if their intersection is not empty.
- Step 3: compute the connected components, which are made of the vertices linked together by edges in the graph G . This allows for the

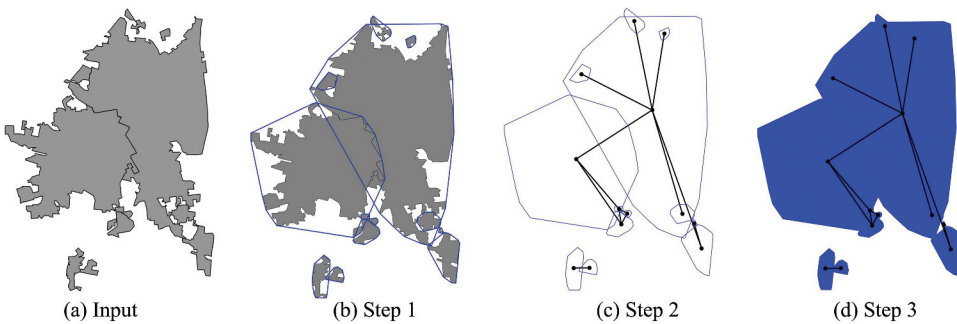


Figure 2. Steps of the city clustering algorithm by intersecting convex envelopes: compute the convex envelope of each built-up area, then compute the intersection of the convex envelopes, and then group the convex envelopes belonging to the same connected component of the intersection graph.

decomposition of the original graph G into connected components, which identify as many cities. A human settlement is a connected component of G , that is, a set of vertices in G that are joined together through a path of edges in G .

- Step 4: each human settlement is associated with its population size.

The algorithm is programmed in Python using the “geopandas” library for data structures and functions (several of them are specialized for treating geospatial data and do not belong to the Python core). It takes one minute of computation time. The steps of the algorithm in terms of graph theory can be implemented in different programming languages, but we present only them in Python:

- (a) Compute the convex envelope of each polygon surrounding the locations using the Python function “.convex_hull;”
- (b) Iteratively merge all pairs of convex envelopes whose intersection is not empty and group all polygons thus obtained into a single set (called “multipolygon”) using the Python function “.unary_union;”
- (c) Extract each polygon from the union of polygons with the function “.geoms.” Each generated polygon is identified as a city. The function “.unary_union” merges all intersecting polygons; the function “.geoms” extracts each polygon from the set of polygons. The result is a set of disjoint polygons;
- (d) The centroid of a polygon is the point $(\bar{x}, \bar{y})$, where $\bar{x}$ is the mean of the abscissae x_i s and $\bar{y}$ the mean of the ordinates y_i s, $i = 1, \dots, n$, of the polygon. Compute the centroids of all polygons constituting the locality with the Python function “.centroid.” Associate the size of the population of its locality with it;
- (e) Assign to each city the population sizes associated with all the centroids it contains, with the spatial join function “.sjoin” from the “geopandas” library. By construction of convexity, all centroids are in one city and one city only. The data are provided by the statistical institutes of each country. For Mexico, the National Institute of Statistics and Geography uses a locality identifier to associate each locality polygon with its population size taken from the census.

4. Results

Table 1 presents descriptive statistics for cities and metropolises within their administrative boundaries and for cities identified by our algorithm. It also presents descriptive statistics for localities. All built-up areas in Mexico are recorded, including those with few inhabitants. We determine

Table 1. Descriptive statistics for Mexican cities in 2020.

Data	Total number of polygons	Population					Total area in km ²
		Total	Maximum	Minimum	Mean	Standard deviation	
Localities	49,948	121,716,575	1,835,486	1	2,437	30,554	38,718
Cities identified by convexification	40,810	121,716,575	19,811,980	1	2,983	108,096	55,254
Administrative cities	401	92,746,167	21,800,320	14,228	231,287	1,178,714	318,948

the threshold $x_{\min}$ at which we begin the distribution of human settlements.

We follow the three steps indicated by Clauset, Shalizi, and Newman (2009) implemented in the Python “powerlaw” library (Alstott, Bullmore, and Plenz, 2014). Table 2 shows the estimated exponent α of Eq. (1) and $x_{\min}$. We first select the values of $x_{\min}$ and the exponent α and use the Kolmogorov–Smirnov statistic. We then validate using a ratio ρ which allows us to compare the log-likelihood of the lognormal distribution with that of the adjusted power-law distribution.

Figure 3 results from our convexification algorithm with $x \geq x_{\min} = 2,126$ inhabitants and the metropolises and cities in their administrative boundaries. It shows that our algorithm also identifies small cities, which is not the case with administrative boundaries.

We compare the set of cities delineated by our convexification to the set of official cities making the “national urban system” (SEDATU-CONAPO-INEGI, 2018b).

The power-law distribution, which the population sizes for settlements of 2,500 inhabitants follow with satisfactory goodness-of-fit, thus remains valid when settlements with less than 2,500 inhabitants are included (column $x_{\min}$ (the estimated minimum value at which the power law begins) in Table 2). The validity of the power law extended to small settlements had never been shown before, to our knowledge. The estimate of $x_{\min} = 2,126$ inhabitants allows us to differentiate between urban and rural locations. The power-law distribution of city size starts at $x_{\min} = 2,126$ inhabitants, which happens to be below the official urban-rural threshold of 2,500. The

Table 2. Power-law distribution fit and goodness-of-fit test for the population sizes of Mexican cities in 2020.

Data	Goodness-of-fit test				
	Cut-off point $x_{\min}$	Distribution function exponent α	Standard deviation	Kolmogorov–Smirnov distance	Clauset, Shalizi, and Newman’s (2009) ratio ρ
Localities	1,918	0.959	0.013	0.008	0.22
Cities identified by convexification	2,126	0.954	0.016	0.010	0.33
Administrative cities	14,228	0.732	0.037	0.040	0.07

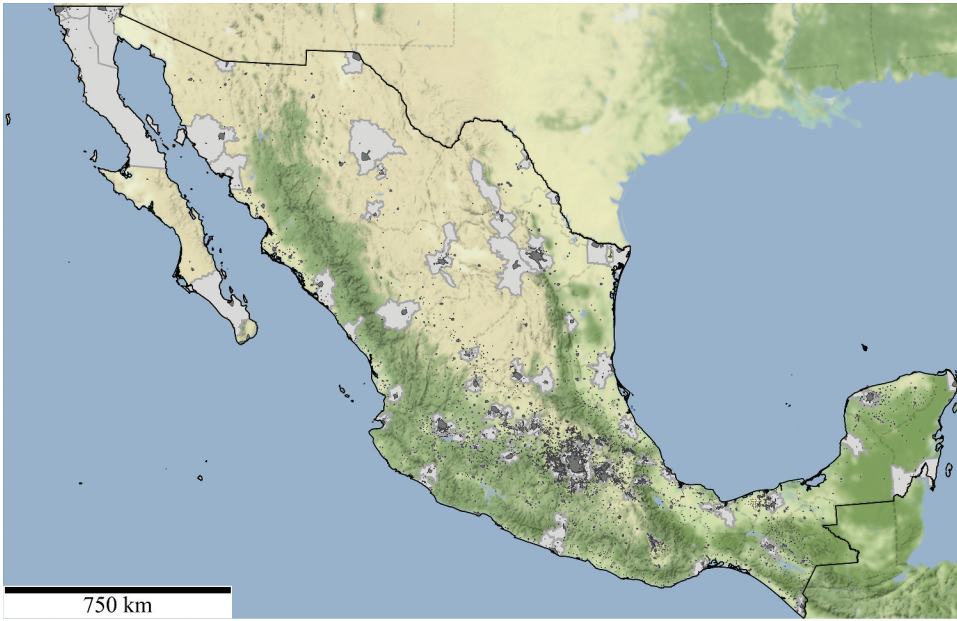


Figure 3. Cities in their administrative boundaries (light gray) and in their boundaries resulting from the convexification algorithm (dark gray) in Mexico.

estimate of $x_{\min}$ could be a good substitute for the 2500 value, as it better reflects the situation. To decide if the fit is good or not, we use a Kolmogorov–Smirnov test with the Kolmogorov–Smirnov distance and the total number of cities.

Table 2 shows that the power law is not rejected at the threshold value of $\rho = 0.10$ for the distribution of population sizes of cities defined by convexification. Clauset, Shalizi, and Newman (2009) rejected the power-law distribution if the ratio ρ is less than 0.10: if ρ is greater than 0.10, then the difference between empirical data and the estimated power-law distribution is due to statistical fluctuations alone; if it is less than 0.10, then the power-law distribution is not plausible for fitting the data. To compute ρ at 2-digit precision, we performed this test with 2,500 sets of simulated data x drawn from a power-law distribution (α given) for $x \geq x_{\min}$ and from sampled empirical data for $x < x_{\min}$. This does not allow us to reject the distribution of population sizes of Mexican cities (delimited with our convexification algorithm) and of localities for ρ at the 10% threshold. The power law is rejected for the distribution of population sizes of cities within their administrative boundaries for the whole of Mexico.

Figure 4 shows the complementary cumulative distribution function $P(x)$ defined in Eq. (3) and the fitted power-law model. Graphically, the power-law distribution fits the distribution of population sizes of cities better than the

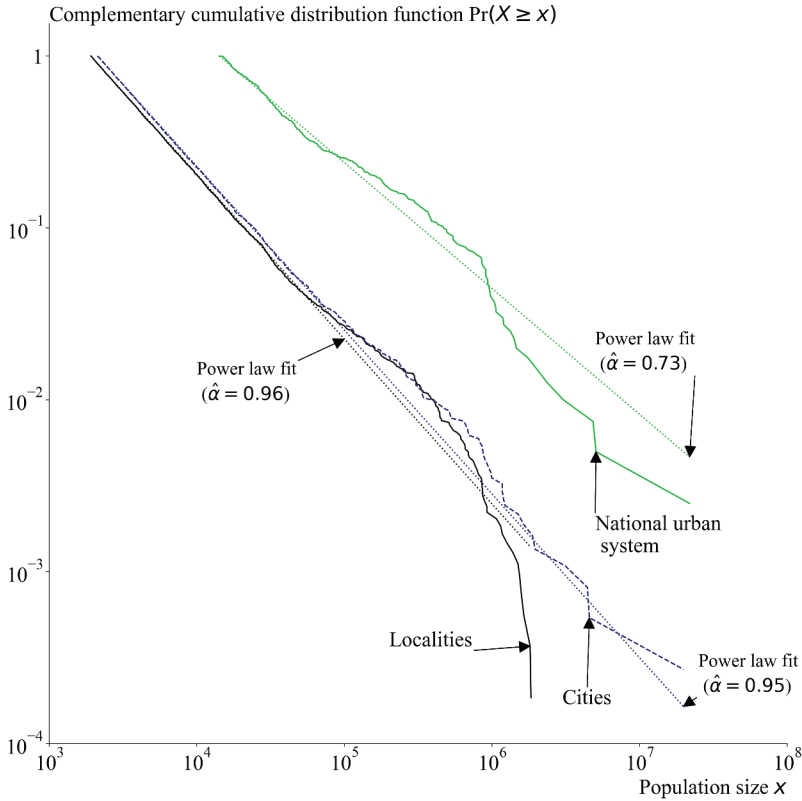


Figure 4. Complementary cumulative distribution function $\Pr(X \geq x)$ above the cutoff point (solid lines) and fitted power law (dotted lines) of the population size of each type of city.

empirical distribution of population sizes of localities and the distribution of administrative cities and metropolises.

Cities and localities have their complementary cumulative distribution function $P(x)$ higher than that expected if the distribution was a power law for cities with population sizes between one hundred thousand and one million inhabitants. This is consistent with the results of Pérez-Campuzano, Guzmán-Vargas, and Angulo-Brown (2015), who, for data between the years 1970 and 2000, found that the population sizes of cities between 10^5 and 10^6 inhabitants exceeded the prediction by a power-law distribution. They attributed this to the accelerated population growth of medium-sized cities relative to large cities and small urban areas.

The power law is not rejected for ρ at the 10% level. To verify that the power-law distribution fits better than the log-normal distribution, the log-likelihood ratio test provides the normalized log-likelihood ratio $\mathcal{R}$. If $\mathcal{R} > 0$, the power-law distribution fits better than the candidate distribution and conversely if it is negative. The power law is not rejected for the distribution of population sizes of cities (according to our convexification algorithm) and

Table 3. Log-likelihood ratio test between power-law and lognormal distributions.

Data	Log-likelihood ratio $\mathcal{R}$	p -value
Localities	1.16	.24
Cities identified by convexification	1.47	.14
Administrative cities	−0.63	.53

localities. The next step in Clauset, Shalizi, and Newman’s (2009) method is to perform the log-likelihood test to determine if the power-law distribution fits better than other distributions, among them the lognormal distribution. The positive value of $\mathcal{R}$ obtained for localities and cities (delimited with our convexification algorithm), in the second column of Table 3, indicates that the power-law distribution provides a better fit than the lognormal distribution. However, the log-likelihood ratio test is not significant at the 5% level (third column of Table 3) and did not allow us to reject the power law. Therefore, the power law fits well for the distribution of population sizes of Mexican cities. The estimated exponent is $\hat{\alpha} = 0.954$ (SD = 0.016) for the city data. Therefore, the population sizes of Mexican cities follow a power-law distribution with α close to 1 and thus verify Zipf’s law, whereas the power-law distribution is rejected for population sizes of Mexican cities in their *administrative* boundaries.

4.1. Cities as a whole

Among the set of cities identified by the algorithm by intersections of convex envelopes of built-up areas, we consider the subset of those where the power law makes sense, that is, for $x \geq x_{\min}$, namely those of 2,126 inhabitants or more, which is the threshold we computed by Clauset, Shalizi, and Newman’s (2009) method (Table 2). It includes 3,705 cities with 10,955 localities.³ Table 4 presents descriptive statistics for this set of cities.

The subset of cities includes 83.1% of the Mexican population, while the population within urban localities includes only 78.8% of the population at the 2020 census. The difference comes from the fact that the algorithm groups

Table 4. Descriptive statistics for Mexican cities ($x \geq x_{\min} = 2,126$ inhabitants) in the year 2020.

Data	Total number of areas	Population					Total area in km ²
		Total	Maximum	Minimum	Mean	Standard deviation	
Cities identified by convexification ($x \geq x_{\min} = 2,126$ inhabitants)	3,705	104,662,608	19,811,980	2,126	28,249	357,816	38,938

³<https://figshare.com/s/bfb32c588b9e12d90cbe>.

localities labeled as rural, whereas these localities are located in the built-up area of the city.

The density of the subset of cities is 2,688 inhabitants per km^2 , while within the administrative boundaries it is only 291 inhabitants per km^2 . This substantial difference comes from the fact that metropolises, which group municipalities, include uninhabited areas. The density of built-up areas in Mexico is 3,143.7 inhabitants per km^2 , which is approximately the value for all cities with a population size greater than $x_{\min} = 2,126$ inhabitants, in contrast to the value of 291 inhabitants for the set of cities within their administrative boundaries. Localities cover exactly the area occupied by built-up areas, but cities in their administrative boundaries also include sparsely populated or uninhabited areas. These areas can be large. The advantage of our convex envelope method is that these areas are minimal. That is why we argue that the subset of cities (for which the population size $x \geq x_{\min} = 2,126$ inhabitants) better represents the built-up areas of cities in Mexico than the set of cities within their administrative boundaries.

This difference is more visible in states with more spatially extensive municipalities. For example, in the states of Baja California Sur and Sonora, the area occupied by administrative cities represents a large proportion of the area of these states (light gray in Figure 5), while the area of cities is much smaller (dark gray in Figure 5). Our identification of cities has the merit of being based on the morphology of built-up areas, regardless of administrative boundaries, which too often mix inhabited and sparsely inhabited or uninhabited areas.

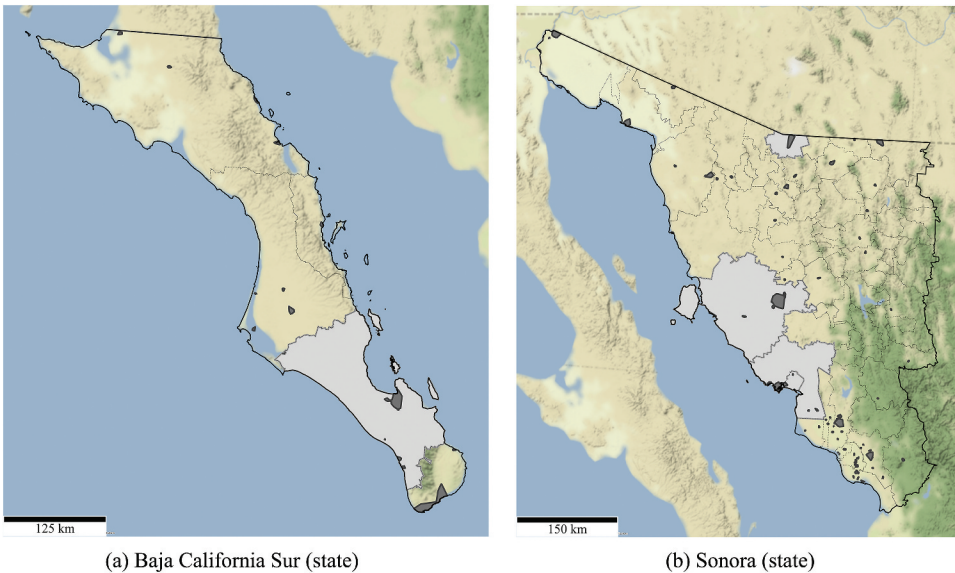


Figure 5. Cities in their administrative boundaries (light gray) and cities delimited by our convexification algorithm (dark gray). Municipal boundaries are indicated in thin black lines.

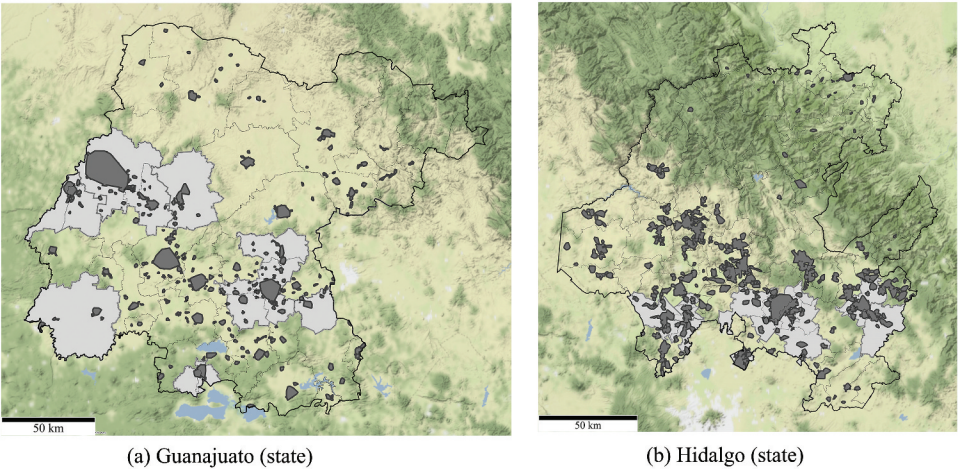


Figure 6. Cities as identified by administrative boundaries (light gray) and cities as identified by our convexification algorithm (dark gray). Municipal boundaries are indicated by thin black lines.

The administrative cities (according to the national urban system) correspond only to large cities and metropolises, while we also identify any urban settlement. For example, in the states of Hidalgo and Guanajuato, important cities, such as Atotonilco el Grande (25,480 inhabitants) and Tepeapa (26,282 inhabitants) in Hidalgo, lie outside metropolises (Figure 6).

Figure 6 shows that many cities, 821, comprise localities belonging to different municipalities, and 92 of them intersect more than just one municipality, such as Mexico City or Puebla in Figure 7. Therefore, we argue, administrative boundaries are inadequate for identifying cities.

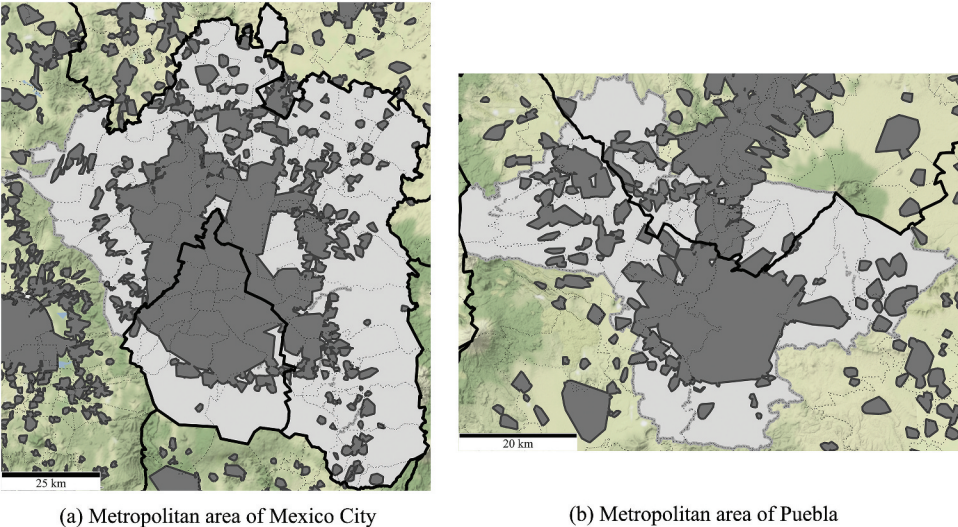


Figure 7. Cities in their administrative boundaries (light gray) and cities delimited by our convexification algorithm (dark gray). Municipal boundaries are indicated in thin black lines.

5. Conclusion

Our method for circumscribing cities is based on the shapes of built-up areas associated with census population size data by block. The algorithm has the advantage of avoiding parameters and overcoming the constraint of administrative boundaries. Instead, the focus is on built-up environments. Using the 2020 census and maps provided by the Mexican National Institute of Statistics and Geography, which also include rural localities with blocks, we have shown that the 2020 population sizes of Mexican cities follow a power-law distribution of exponent $\alpha = 0.954$ (SD = 0.016) with $x_{\min} = 2,126$ inhabitants. We therefore extended the power-law distribution to the smallest recorded settlements.

The power law obviously depends on the definition of cities. The power law is rejected for metropolises and cities within their administrative boundaries, but we showed that it is valid within the boundaries identified by the convexification of building areas. The algorithm for clustering cities by the intersection of convex envelopes is an alternative to the adjacency relation used so far in city clustering (Rozenfeld et al., 2008). It allows built-up areas that are close enough, but not necessarily contiguous, to be grouped into a single set that we identify as a city. The algorithm also avoids considering commuting flows.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

Maravillo Gomez was supported by a CONACYT scholarship.

Data availability statement

The site <https://figshare.com/s/bfb32c588b9e12d90cbe> provides three files in Python (released under the General Public License v3): to prepare the data, compute the convex envelopes and their intersections, test whether the population sizes of cities follow a power-law distribution, and map cities at the national and state levels. The site contains the coordinates of the polygons and their associated population sizes. The data come from the following resources available in the public domain:

- Urban and rural locality boundaries with blocks, states, and municipalities from Mexico's National Geostatistical Framework 2020 (INEGI, 2020b).
- Data from the Mexican Census of Population and Housing 2020 (INEGI, 2020a).
- Data on the boundaries and population size of cities and metropolises of the national urban system in 2018 (SEDATU-CONAPO, 2018a).

References

- Alstott, J., Bullmore, E., and Plenz, D. (2014). Powerlaw: a Python package for analysis of heavy-tailed distributions. *PLoS One*, 9(1): e85777. doi:[10.1371/journal.pone.0085777](https://doi.org/10.1371/journal.pone.0085777).
- Auerbach, F. (1913). Das Gesetz der Bevölkerungskonzentration. *Petermanns Geographische Mitteilungen*, 59: 74–76.
- Bee, M., Riccaboni, M., and Schiavo, S. (2019). Distribution of city size: Gibrat, Pareto, Zipf. In L. D'Acci (Ed.), *The Mathematics of Urban Morphology*. Cham, Switzerland: Birkhäuser, 77–91.
- Bergs, R. (2018). The detection of natural cities in the Netherlands – nocturnal satellite imagery and Zipf's law. *Review of Regional Research*, 38(2): 111–140. doi:[10.1007/s10037-018-0122-6](https://doi.org/10.1007/s10037-018-0122-6).
- Berry, B. J. and Okulicz-Kozaryn, A. (2012). The city size distribution debate: resolution for U.S. urban regions and megalopolitan areas. *Cities*, 29: S17–23. doi:[10.1016/j.cities.2011.11.007](https://doi.org/10.1016/j.cities.2011.11.007).
- Cao, W., Dong, L., Wu, L., and Liu, Y. (2020). Quantifying urban areas with multi-source data based on percolation theory. *Remote Sensing of Environment*, 241: 111730. doi:[10.1016/j.rse.2020.111730](https://doi.org/10.1016/j.rse.2020.111730).
- Chen, Y., Wang, J., Long, Y., Zhang, X., and Li, X. (2022). Defining urban boundaries by characteristic scales. *Computers, Environment and Urban Systems*, 94: 101799. doi:[10.1016/j.compenvurbsys.2022.101799](https://doi.org/10.1016/j.compenvurbsys.2022.101799).
- Clauset, A., Shalizi, C. R., and Newman, M. E. (2009). Power-law distributions in empirical data. *SIAM Review*, 51(4): 661–703. doi:[10.1137/070710111](https://doi.org/10.1137/070710111).
- Cristelli, M., Batty, M., and Pietronero, L. (2012). There is more than a power law in Zipf. *Scientific Reports*, 2: 1–7. doi:[10.1038/srep00812](https://doi.org/10.1038/srep00812).
- Dijkstra, L. and Poelman, H. (2012). Cities in Europe: the new OECD-EC definition. *Regional Focus*, 1: 1–13.
- Dijkstra, L., Poelman, H., and Veneri, P. (2019). The EU-OECD definition of a functional urban area. *OECD Regional Development Working Papers*, 11: 1–18.
- Garmestani, A. S., Allen, C. R., and Gallagher, C. M. (2008). Power laws, discontinuities and regional city size distributions. *Journal of Economic Behavior & Organization*, 68(1): 209–216. doi:[10.1016/j.jebo.2008.03.011](https://doi.org/10.1016/j.jebo.2008.03.011).
- Gillespie, C. S. (2015). Fitting heavy tailed distributions: the powerLaw package. *Journal of Statistical Software*, 64(2): 1–16. doi:[10.18637/jss.v064.i02](https://doi.org/10.18637/jss.v064.i02).
- González-Val, R. (2010). The evolution of US city size distribution from a long-term perspective (1900–2000). *Journal of Regional Science*, 50(5): 952–972. doi:[10.1111/j.1467-9787.2010.00685.x](https://doi.org/10.1111/j.1467-9787.2010.00685.x).
- Holmes, T. J. and Lee, S. (2010). Cities as six-by-six-mile squares: Zipf's law? In E. L. Glaeser (Ed.), *Agglomeration Economics*. Chicago: University of Chicago Press. chapter 3, 105–131.
- Hsu, W. -T. (2012). Central place theory and city size distribution. *The Economic Journal*, 122(563): 903–932. doi:[10.1111/j.1468-0297.2012.02518.x](https://doi.org/10.1111/j.1468-0297.2012.02518.x).
- INEGI. (2020a). Data from: Censo de Población y Vivienda 2020. Principales resultados por localidad (ITER) [dataset]. https://www.inegi.org.mx/programas/ccpv/2020/default.html#Datos_abiertos, Access date: August 30, 2021.
- INEGI. (2020b). Data from: Marco Geoestadístico. Censo de Población y Vivienda 2020 [dataset]. <https://www.inegi.org.mx/app/biblioteca/ficha.html?upc=889463807469>, Access date: August 30, 2021.
- Jiang, B. and Jia, T. (2011). Zipf's law for all the natural cities in the United States: a geospatial perspective. *International Journal of Geographical Information Science*, 25(8): 1269–1281. doi:[10.1080/13658816.2010.510801](https://doi.org/10.1080/13658816.2010.510801).

- Mandelbrot, B. (1966). Information theory and psycholinguistics: a theory of word frequencies. In P. F. Lazarsfeld and N. W. Henry (Eds.), *Readings in Mathematical Social Sciences* (pp. 350–368). Cambridge, Massachusetts: MIT Press.
- Masucci, P. (2015). On the problem of boundaries and scaling for urban street networks. *Journal of the Royal Society Interface*, 12: 1–7. doi:10.1098/rsif.2015.0763.
- Matthews, S. A. (2020). Methods and applications in spatial demography: 2. *Mathematical Population Studies*, 27(1): 1–7. doi:10.1080/08898480.2020.1715123.
- Montero, G., Tannier, C., and Thomas, I. (2021). Delineation of cities based on scaling properties of urban patterns: a comparison of three methods. *International Journal of Geographical Information Science*, 35(5): 919–947. doi:10.1080/13658816.2020.1817462.
- Moreno-Monroy, A. I., Schiavina, M., and Veneri, P. (2021). Metropolises in the world. Delineation and population trends. *Journal of Urban Economics*, 125: 103242. doi:10.1016/j.jue.2020.103242.
- Newman, M. E. (2005). Power laws, pareto distributions and Zipf's law. *Contemporary Physics*, 46(5): 323–351. doi:10.1080/00107510500052444.
- Oliveira, E. A., Furtado, V., Andrade, J. S., and Makse, H. A. (2018). A worldwide model for boundaries of urban settlements. *Royal Society Open Science*, 5(5): 180468. doi:10.1098/rsos.180468.
- Pérez-Campuzano, E., Guzmán-Vargas, L., and Angulo-Brown, F. (2015). Distributions of city sizes in Mexico during the 20th century. *Chaos, Solitons, and Fractals*, 73: 64–70. doi:10.1016/j.chaos.2014.12.015.
- Puente-Ajovín, M., Ramos, A., and Sanz-Gracia, F. (2020). Is there a universal parametric city size distribution? Empirical evidence for 70 countries. *The Annals of Regional Science*, 65(3): 727–741. doi:10.1007/s00168-020-01001-6.
- Ren, Z., Jiang, B., and Seipel, S. (2019). Capturing and characterizing human activities using building locations in America. *ISPRS International Journal of Geo-Information*, 8(5): 200. doi:10.3390/ijgi8050200.
- Rosen, K. T. and Resnick, M. (1980). The size distribution of cities: an examination of the Pareto law and primacy. *Journal of Urban Economics*, 8(2): 165–186. doi:10.1016/0094-1190(80)90043-1.
- Rozenfeld, H. D., Rybski, D., Andrade, J. S., Batty, M., Stanley, H. E., and Makse, H. A. (2008). Laws of population growth. *Proceedings of the National Academy of Sciences*, 105(48): 18702–18707. doi:10.1073/pnas.0807435105.
- Rozenfeld, H. D., Rybski, D., Gabaix, X., and Makse, H. A. (2011). The area and population of cities: new insights from a different perspective on cities. *The American Economic Review*, 101(5): 2205–2225. doi:10.1257/aer.101.5.2205.
- Schmidheiny, K. and Suedekum, J. (2015). The pan-European population distribution across consistently defined functional urban areas. *Economics Letters*, 133: 10–13. doi:10.1016/j.econlet.2015.05.013.
- SEDATU-CONAPO. (2018a). Mapas del SUN 2018 con los municipios y localidades que lo integran [dataset]. http://www.conapo.gob.mx/work/models/CONAPO/Marginacion/Datos_Abiertos/SUN/Mapa_SUN_2018.rar, Access date: August 30, 2021.
- SEDATU-CONAPO-INEGI. (2018b). Delimitación de las Zonas Metropolitanas de México 2015 [online]. https://www.gob.mx/cms/uploads/attachment/file/344506/1_Preliminares_hasta_V_correcciones_11_de_julio.pdf, Access date: July 6, 2021.
- Soo, K. T. (2005). Zipf's law for cities: a cross-country investigation. *Regional Science and Urban Economics*, 35(3): 239–263. doi:10.1016/j.regsciurbeco.2004.04.004.
- Tannier, C. and Thomas, I. (2013). Defining and characterizing urban boundaries: a fractal analysis of theoretical cities and Belgian cities. *Computers, Environment and Urban Systems*, 41: 234–248. doi:10.1016/j.compenvurbsys.2013.07.003.

- Tannier, C., Thomas, I., Vuidel, G., and Frankhauser, P. (2011). A fractal approach to identifying urban boundaries. *Geographical Analysis*, 43(2): 211–227.
- Urzúa, C. M. (2001). Las ciudades mexicanas no siguen la ley de Zipf. *Estudios Demográficos y Urbanos*, 16(3): 661–669. doi:[10.24201/edu.v16i3.1118](https://doi.org/10.24201/edu.v16i3.1118).
- Veneri, P. (2016). City size distribution across the OECD: does the definition of cities matter? *Computers, Environment and Urban Systems*, 59: 86–94. doi:[10.1016/j.compenvurbsys.2016.05.007](https://doi.org/10.1016/j.compenvurbsys.2016.05.007).
- Verbavatz, V. and Barthelemy, M. (2020). The growth equation of cities. *Nature*, 587: 397–401. doi:[10.1038/s41586-020-2900-x](https://doi.org/10.1038/s41586-020-2900-x).
- Wang, C., Yu, B., Chen, Z., Liu, Y., Song, W., Li, X., Yang, C., Small, C., Shu, S., and Wu, J. (2022). Evolution of urban spatial clusters in China: a graph-based method using nighttime light data. *Annals of the American Association of Geographers*, 112(1): 56–77. doi:[10.1080/24694452.2021.1914538](https://doi.org/10.1080/24694452.2021.1914538).
- Wu, W., Zhao, H., and Jiang, S. (2018). A Zipf's law-based method for mapping urban areas using NPP-VIIRS nighttime light data. *Remote Sensing*, 10(1): 130. doi:[10.3390/rs10010130](https://doi.org/10.3390/rs10010130).
- Xu, Z., Jiao, L., Lan, T., Zhou, Z., Cui, H., Li, C., Xu, G., and Liu, Y. (2021). Mapping hierarchical urban boundaries for global urban settlements. *International Journal of Applied Earth Observation and Geoinformation*, 103: 102480. doi:[10.1016/j.jag.2021.102480](https://doi.org/10.1016/j.jag.2021.102480).
- Yin, J., Soliman, A., Yin, D., and Wang, S. (2017). Depicting urban boundaries from a mobility network of spatial interactions: a case study of Great Britain with geo-located Twitter data. *International Journal of Geographical Information Science*, 31(7): 1293–1313. doi:[10.1080/13658816.2017.1282615](https://doi.org/10.1080/13658816.2017.1282615).
- Yu, B., Shu, S., Liu, H., Song, W., Wu, J., Wang, L., and Chen, Z. (2014). Object-based spatial cluster analysis of urban landscape pattern using nighttime light satellite images: a case study of China. *International Journal of Geographical Information Science*, 28(11): 2328–2355. doi:[10.1080/13658816.2014.922186](https://doi.org/10.1080/13658816.2014.922186).
- Zhou, Y., Smith, S. J., Zhao, K., Imhoff, M., Thomson, A., Bond-Lamberty, B., Asrar, G. R., Zhang, X., He, C., and Elvidge, C. D. (2015). A global map of urban extent from nightlights. *Environmental Research Letters*, 10(5): 054011. doi:[10.1088/1748-9326/10/5/054011](https://doi.org/10.1088/1748-9326/10/5/054011).
- Zipf, G. K. (1949). *Human Behavior and the Principle of Least Effort: An Introduction to Human Ecology*. Cambridge, Massachusetts: Addison-Wesley Press.

Copyright of Mathematical Population Studies is the property of Routledge and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.